Linear Equations & Lines

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the system of linear inequalities represented by the shaded region in the graph below. The boundary lines are given by $y = -2x + 10$ and $y = x + 1$. The point $(k, 4)$ lies on the boundary line $y = -2x + 10$ and is within the feasible region. Determine the value of k.

The point $(k, 4)$ lies on the line $y = -2x + 10$. Substituting the coordinates of the point into the equation of the line, we get $4 = -2k + 10$. Solving for k: $2k = 10 - 4$ $2k = 6$ $k = 3$. Now we must verify that the point $(3, 4)$ is within the feasible region defined by the system of inequalities. The inequalities corresponding to the boundary lines $y = -2x + 10$ and $y = x + 1$ and the shaded region are: 1. $y \leq -2x + 10$ (since the shaded region is below or on this line, including the point (5,0) as.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the system of linear equations graphed below: $\{2x + y = 4; x - y = -1$ What is the solution (x, y) to this system?

The solution to a system of linear equations is the point where the two lines intersect. By examining the graph, we can see that the intersection point is at $(1, 2)$. We can verify this algebraically by substituting the point into both equations: For the first equation: $2(1) + 2 = 2 + 2 = 4$. This is true. For the second equation: $1 - 2 = -1$. This is true. Therefore, the solution to the system is $(1, 2)$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the system of linear equations: $\{2x + y = 7; x - y = 2$ What is the value of x?

We are given the system of linear equations: 1) $2x + y = 7$ 2) $x - y = 2$ To find the value of x, we can add the two equations together to eliminate y: $(2x + y) + (x - y) = 7 + 2$ $2x + x + y - y = 9$ $3x = 9$ Now, divide by 3 to solve for x: $x = \frac{9}{3}$ $x = 3$ To verify, we can substitute $x = 3$ into the second equation to find y: $3 - y = 2$ $-y = 2 - 3$ $-y = -1$ $y = 1$ Now substitute $x = 3$ and $y = 1$ into the first equation: $2(3) + 1 = 6 + 1 = 7$, which is correct. Therefore, the value of x.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear equations: $\{2x + y = 5; x - y = 1$ What is the solution (x, y) for this system?

We can solve this system using the elimination method. Add the two equations together: $(2x + y) + (x - y) = 5 + 1$ $3x = 6$ Divide by 3 to find x: $x = \frac{6}{3}$ $x = 2$ Now substitute the value of x back into either of the original equations to solve for y. Using the first equation: $2(2) + y = 5$ $4 + y = 5$ Subtract 4 from both sides: $y = 5 - 4$ $y = 1$ Therefore, the solution is $(2, 1)$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let the rational function be defined as $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$. If the graph of $y = f(x)$ has a removable discontinuity (a hole) at $x = 2$, determine the value of the parameter a.

For the rational function $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$ to have a removable discontinuity at $x = 2$, the factor $(x - 2)$ must be present in both the numerator and the denominator, and it must cancel out. First, factor the denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$. For $(x - 2)$ to be a factor of the numerator $x^2 - ax + 6$, substituting $x = 2$ into the numerator must result in zero: $(2)^2 - a(2) + 6 = 0$. This simplifies to $4 - 2a + 6 = 0$, which gives $10 - 2a = 0$. Solving for a, we get $2a = 10$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 4x + 7$. The graph of $y = f(x)$ is subjected to a sequence of transformations. First, it is translated by a units horizontally and b units vertically. Let this transformed function be $g(x)$. Subsequently, the graph of $y = g(x)$ is reflected in the x-axis to produce the graph of $y = h(x)$. If the vertex of the graph of $y = h(x)$ is at $(-2, -3)$, determine the value of $a^2 + b^2$.

The original function is $f(x) = x^2 - 4x + 7$. We can complete the square to find the vertex: $f(x) = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$. The vertex of $y = f(x)$ is at $(2, 3)$. When $y = f(x)$ is translated by a units horizontally and b units vertically, the new function $g(x)$ is given by $g(x) = f(x - a) + b$. The vertex of $g(x)$ will be at $(2 + a, 3 + b)$. Next, the graph of $y = g(x)$ is reflected in the x-axis to produce $y = h(x)$. This means $h(x) = -g(x)$. The vertex of $h(x)$ will be at $(2 + a, -(3 + b))$. We are g.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the linear equation $y = mx + c$ is shown below. The line passes through the point $(-3, -11)$ and intersects the y-axis at $(0, k)$. If the slope of the line is -3, determine the value of k.

The equation of a line is given by $y = mx + c$, where m is the slope and c is the y-intercept. We are given that the slope $m = -3$. Therefore, the equation of the line is $y = -3x + c$. The line passes through the point $(-3, -11)$. We can substitute these coordinates into the equation to find c: $-11 = -3(-3) + c$ $-11 = 9 + c$ $c = -11 - 9$ $c = -20$ The y-intercept is the point where the line crosses the y-axis, which is $(0, k)$. For the equation $y = -3x + c$, when $x = 0$, $y = c$. Thus, $k = c$. Since.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

What is the solution (x, y) to the system of equations below? $x + y, = 8; 2x - y, = 7$

To solve the system of linear equations, we can use the elimination method by adding the two equations together: $(x + y) + (2x - y) = 8 + 7$ Combine like terms: $3x = 15 \Rightarrow x = 5$ Substitute $x = 5$ into the first equation to find y: $5 + y = 8 \Rightarrow y = 3$ Thus, the solution to the system is $(5, 3)$. - $(3, 5)$ results from reversing the x- and y-coordinates. - $(4, 1)$ satisfies the second equation $(2(4) - 1 = 7)$ but fails the first ($4 + 1 = 5eq8$). - $(6, 2)$ satisfies the first equat.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the polynomial $P(x) = x^3 - 13x + 12$. The graph of $y = P(x)$ is shown below. The polynomial can be factored into the form $P(x) = (x - a)(x - b)(x - c)$ where $a < b < c$. Determine the value of $c - a$.

The problem requires factoring a cubic polynomial and interpreting the roots from a graph to solve for a difference between roots. First, we identify possible rational roots of $P(x) = x^3 - 13x + 12$ using the Rational Root Theorem. Possible rational roots are divisors of the constant term (12) divided by divisors of the leading coefficient (1), which are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$. From the graph, we can visually estimate the x-intercepts (roots) of the polynomial. The graph shows roots at.

10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the function $f(x) = ax^2 + b$ is shown below. The graph passes through the points $(-2, 13)$ and $(3, 28)$. Determine the value of b.

The function is given by $f(x) = ax^2 + b$. We are given two points that lie on the graph of this function: $(-2, 13)$ and $(3, 28)$. Substitute the first point into the equation: $13 = a(-2)^2 + b$, which simplifies to $13 = 4a + b$. (Equation 1) Substitute the second point into the equation: $28 = a(3)^2 + b$, which simplifies to $28 = 9a + b$. (Equation 2) Now we have a system of two linear equations with two variables, a and b: 1) $4a + b = 13$ 2) $9a + b = 28$ To find the value of b, we can eliminate a.

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11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar graph shows the frequency distribution of scores obtained by 20 students on a 5-point quiz, where no student originally scored 5. A group of k additional students take the quiz later, and each of these students earns a score of 5. If the mean score of the combined group of $20 + k$ students is equal to 3, what is the value of k?

Step 1: Calculate the total sum of the scores for the original 20 students from the bar graph: $\text{Sum}_{\text{original}} = (1 \times 4) + (2 \times 7) + (3 \times 6) + (4 \times 3) = 4 + 14 + 18 + 12 = 48$ Step 2: Add the k additional students, each scoring 5 points. The new sum of scores is $48 + 5k$, and the new total number of students is $20 + k$. Step 3: Set up the equation for the new mean: $\frac{48 + 5k}{20 + k} = 3$ Step 4: Solve for k: $48 + 5k = 3(20 + k)$ $48 + 5k = 60 + 3k$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = x^2$ is shown below. It is transformed to the graph of $g(x) = (x - 3)^2 + 2$. Which transformation maps $f(x)$ to $g(x)$?

The function $g(x) = (x - 3)^2 + 2$ is a transformation of the base function $f(x) = x^2$. In the general form of a quadratic transformation, $h(x) = a(x - h)^2 + k$, a value of h represents a horizontal translation, and a value of k represents a vertical translation. If h is positive, the translation is to the right. If h is negative, the translation is to the left. If k is positive, the translation is upwards. If k is negative, the translation is downwards. In $g(x) = (x - 3)^2 + 2$, we have $h = 3$ and.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = a \cdot b^x + c$ is shown below. The graph passes through the points $(0, 1)$ and $(1, 7)$, and has a horizontal asymptote at $y = -5$. Determine the values of a and b.

The function is given by $f(x) = a \cdot b^x + c$. The horizontal asymptote is at $y = -5$, which means $c = -5$. So, the function is $f(x) = a \cdot b^x - 5$. We are given that the graph passes through the point $(0, 1)$. Substituting these coordinates into the function: $1 = a \cdot b^0 - 5$ $1 = a \cdot 1 - 5$ $1 = a - 5$ $a = 6$ Now we know $a = 6$, so the function is $f(x) = 6 \cdot b^x - 5$. We are also given that the graph passes through the point $(1, 7)$. Substituting these coordinates into the function.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(1, 5)$ and the parabola passes through the point $(3, -3)$. Determine the value of a.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. \\ Given the vertex is $(1, 5)$, the function can be written as $f(x) = a(x - 1)^2 + 5$. \\ The graph also passes through the point $(3, -3)$. We can substitute these coordinates into the equation to solve for a: \\ $-3 = a(3 - 1)^2 + 5$ \\ $-3 = a(2)^2 + 5$ \\ $-3 = 4a + 5$ \\ Subtract 5 from both sides: \\ $-3 - 5 = 4a$ \\ $-8 = 4a$ \\ Divide by 4: \\ $a = \frac{-8}{4}$ \\ $a = -2$. \\

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Functions & Transformations**

The graph of the quadratic function $f(x) = ax^2 + bx - 5$ is shown below. The function has roots at $x = -5$ and $x = 1$. Find the value of a.

Since the roots of the quadratic function $f(x) = ax^2 + bx - 5$ are $x = -5$ and $x = 1$, we can write the factored form of the quadratic as $f(x) = a(x - (-5))(x - 1) = a(x + 5)(x - 1)$. Expanding this, we get $f(x) = a(x^2 - x + 5x - 5) = a(x^2 + 4x - 5) = ax^2 + 4ax - 5a$. We are given that the constant term is -5. Therefore, $-5a = -5$, which implies $a = 1$. The quadratic function is $f(x) = 1x^2 + 4(1)x - 5 = x^2 + 4x - 5$. This is consistent with the given information.

16 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions**

The bar chart below shows the number of students who participated in different sports at a school. What is the total number of students who participated in basketball and volleyball?

The bar chart shows that 22 students participated in basketball and 18 students participated in volleyball. To find the total number of students who participated in these two sports, we add the number of participants for each sport: 22 (basketball) + 18 (volleyball) = 40 students.

17 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities**

The following bar chart shows the number of students who achieved different scores on a recent quiz. What is the mode of the quiz scores?

The mode of a dataset is the value that appears most frequently. In the given bar chart, the score of 80 has the highest frequency, with 25 students achieving this score. Therefore, the mode of the quiz scores is 80.

18 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series**

The bar chart below shows the number of students in four different clubs at a school. What is the total number of students in the Science Club and the Chess Club combined?

From the bar chart, the number of students in the Science Club is 18 and the number of students in the Chess Club is 12. The total number of students in both clubs is the sum of these two values: $18 + 12 = 30$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Let S be the set of all distinct real solutions to the equation $(x^2 - x + 1)^4 - 10x^2(x^2 - x + 1)^2 + 9x^4 = 0$. What is the sum of the squares of the elements of S?

To solve the equation without expanding the degree-8 polynomial, let $u = x^2 - x + 1$. Substituting u into the equation yields a quadratic form in u^2: $u^4 - 10x^2u^2 + 9x^4 = 0$. Factoring this as a quadratic in u^2: $(u^2 - x^2)(u^2 - 9x^2) = 0$. Applying the difference of squares factorization to each factor gives: $(u - x)(u + x)(u - 3x)(u + 3x) = 0$. Now substitute $u = x^2 - x + 1$ back into each linear factor in u: 1. $u - x = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0$, giving the re.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Radicals & Exponents

The rational function $R(x) = \frac{x^2 - 7x + k}{x - 3}$ has a removable discontinuity (a hole) at $x = 3$. For what value of k does this occur?

A rational function of the form $R(x) = \frac{P(x)}{Q(x)}$ has a removable discontinuity (a hole) at $x = a$ if $(x - a)$ is a factor of both the numerator $P(x)$ and the denominator $Q(x)$. In this case, the denominator is $Q(x) = x - 3$, so the discontinuity occurs at $x = 3$. For this discontinuity to be removable, $(x - 3)$ must also be a factor of the numerator, $P(x) = x^2 - 7x + k$. This means that when $x = 3$, $P(3)$ must equal zero. We can substitute $x = 3$ into the numerator and set the expression equal t.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the xy-plane, line ℓ_1 has the equation $3x + 4y = 24$. Line ℓ_2 is perpendicular to ℓ_1 and intersects ℓ_1 at a point P in the first quadrant. If line ℓ_1, line ℓ_2, and the x-axis enclose a triangle with an area of 6, what is the x-coordinate of P?

a. $\frac{12}{5}$

b. 3

c. $\frac{24}{5}$ ✓

d. $\frac{32}{5}$

Let $P = (x_0, y_0)$ be the intersection point of lines ℓ_1 and ℓ_2. Since P lies in the first quadrant, $x_0 > 0$ and $y_0 > 0$. 1. **Find the x-intercept of ℓ_1:** Setting $y = 0$ in $3x + 4y = 24$ yields $3x = 24 \Rightarrow x = 8$. Thus, ℓ_1 intersects the x-axis at $A(8, 0)$. 2. **Find the x-intercept of ℓ_2:** The slope of ℓ_1 is $-\frac{3}{4}$. Since $\ell_2 \perp \ell_1$, the slope of ℓ_2 is the negative reciprocal, $m_2 = \frac{4}{3}$. The equation of ℓ_2 pas.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a triangle ABC. The line segment AD is the angle bisector of angle BAC, where D lies on BC. Points E and F are on AC and AB respectively, such that EF is parallel to BC. Given that AB = 10 units, AC = 15 units, BD = 6 units, and the area of triangle AEF is $\frac{4}{9}$ of the area of triangle ABC. Determine the length of CD in units.

a. 7.5

b. **9** ✓

c. 12

d. 15

Let CD = x. By the Angle Bisector Theorem, $\frac{AB}{AC} = \frac{BD}{CD}$. Substituting the given values, we have $\frac{10}{15} = \frac{6}{x}$. This simplifies to $10x = 15 \times 6$, so $10x = 90$, which gives $x = 9$. Thus, CD = 9 units. Since EF is parallel to BC, triangle AEF is similar to triangle ABC. The ratio of the areas of two similar triangles is the square of the ratio of their corresponding sides. Given that the area of triangle AEF is $\frac{4}{9}$ of the area of triangle ABC, we have.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A farmer is planning to fence a rectangular plot of land adjacent to a straight river. The farmer has 100 meters of fencing material. The side of the plot along the river does not require fencing. Let the dimensions of the rectangular plot be x meters (perpendicular to the river) and y meters (parallel to the river). The total area of the plot is given by $A = xy$. If the farmer wants to maximize the area, what are the dimensions x and y that achieve this maximum area?

a. $x = 25$ **m,** $y = 50$ **m** ✓

b. $x = 30$ m, $y = 40$ m

c. $x = 20$ m, $y = 60$ m

d. $x = 35$ m, $y = 30$ m

The farmer has 100 meters of fencing material. The fencing will be used for three sides of the rectangular plot: two sides of length x (perpendicular to the river) and one side of length y (parallel to the river). The side along the river does not need fencing. Therefore, the constraint on the fencing material is given by the equation: $2x + y = 100$ The area of the rectangular plot is given by $A = xy$. We need to express the area A as a function of a single variable. From the constraint equation, we can exp.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the xy-plane, line ℓ passes through the points $(-2, 5)$ and $(4, -1)$. Line k is perpendicular to line ℓ and has the equation $ax + 2y = c$, where a and c are constants. If line k passes through the point $(3, 1)$, what is the value of c?

a. -4 ✓

b. 2

c. 4

d. 8

First, find the slope of line ℓ using the slope formula: $m_\ell = \frac{-1-5}{4-(-2)} = \frac{-6}{6} = -1$ Since line k is perpendicular to line ℓ, its slope m_k is the negative reciprocal of m_ℓ: $m_k = -\frac{1}{-1} = 1$ Next, rewrite the standard form equation of line k, $ax + 2y = c$, in slope-intercept form to find the relationship between a and the slope: $2y = -ax + c \Rightarrow y = -\frac{a}{2}x + \frac{c}{2}$ The slope is $-\frac{a}{2}$. Setting this equal to 1.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider a line that passes through the point $(2, 5)$ and has a slope of -3. If the equation of this line is written in the form $y = mx + c$, and it is also known that the point $(k, -7)$ lies on this line, determine the value of k.

a. k = 5

b. **k = 6 ✓**

c. k = 7

d. k = 8

The equation of a line with slope m passing through a point (x_1, y_1) can be found using the point-slope form: $y - y_1 = m(x - x_1)$. Given the point $(2, 5)$ and slope $m = -3$, we have: $y - 5 = -3(x - 2)$ Expand the equation: $y - 5 = -3x + 6$ Rearrange into the form $y = mx + c$: $y = -3x + 6 + 5$ $y = -3x + 11$ So, the equation of the line is $y = -3x + 11$. The value of c is 11. We are also given that the point $(k, -7)$ lies on this line. Substitute these coordinates into the equation.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a line passing through the points $A(2, k)$ and $B(6, 10)$. If the slope of the line segment AB is m and $m = \frac{1}{2}$, find the value of k.

a. $k = 6$ ✓

b. $k = 8$

c. $k = 12$

d. $k = 4$

The slope of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $A(2, k)$ and $B(6, 10)$, we have $x_1 = 2$, $y_1 = k$, $x_2 = 6$, and $y_2 = 10$. The slope is given as $m = \frac{1}{2}$. Substituting the values into the slope formula: $\frac{1}{2} = \frac{10 - k}{6 - 2}$ $\frac{1}{2} = \frac{10 - k}{4}$ To solve for k, we can cross-multiply: $1 \times 4 = 2 \times (10 - k)$ $4 = 20 - 2k$ Now, isolate the term with k.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

In the xy-plane, the graph of the linear equation $4x - 7y = 28$ intersects the x-axis at the point $(c, 0)$. What is the value of c?

a. 7 ✓

b. -4

c. 4

d. 28

To find the coordinates of the x-intercept, substitute $y = 0$ into the equation $4x - 7y = 28$: $4(c) - 7(0) = 28$ $4c = 28$ $c = 7$ - Choice A is correct. - Choice B (-4) is the result of solving for the y-intercept by setting $x = 0$, yielding $-7y = 28 \Rightarrow y = -4$. - Choice C (4) confuses the coefficient of x with the value of the intercept or computes $\frac{28}{7}$. - Choice D (28) incorrectly takes the constant term as the intercept.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Factoring Polynomials

A line passes through the points $(2, 3)$ and $(6, 11)$. What is the slope of this line?

a. 2 ✓

b. $\frac{1}{2}$

c. -2

d. $-\frac{1}{2}$

The slope of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $(2, 3)$ and $(6, 11)$, let $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (6, 11)$. Substituting these values into the slope formula: $m = \frac{11 - 3}{6 - 2}$ $m = \frac{8}{4}$ $m = 2$ Therefore, the slope of the line is 2.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Radicals & Exponents

Consider the line with the equation $y = 2x + 3$. What is the y-intercept of this line?

a. **3** ✓

b. 2

c. $-3/2$

d. $-2/3$

The equation of a line in slope-intercept form is given by $y = mx + c$, where m is the slope and c is the y-intercept. In the given equation, $y = 2x + 3$, we can identify the slope $m = 2$ and the y-intercept $c = 3$. Therefore, the y-intercept is 3.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Rational Expressions

A line has slope -2 and passes through $(-6, 3)$. What is its y-intercept?

a. -11

b. **-9 ✓**

c. 9

d. -8

$y - 3 = -2(x + 6) \Rightarrow y = -2x - 9$, so the y-intercept is -9.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Statistics & Data Analysis

What is the slope of the line through $(1, -1)$ and $(3, 7)$?

a. 3

b. **4 ✓**

c. $\frac{1}{4}$

d. -4

$m = \frac{7-(-1)}{3-(1)} = \frac{8}{2} = 4$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratic Functions

Consider the system of linear equations: $kx + 12y = 2k + 6$ $3x + ky = 9$ where k is a constant. For which value of k does the system have no solution?

a. -6 ✓

b. -2

c. 2

d. 6

A system of two linear equations $A_1x + B_1y = C_1$ and $A_2x + B_2y = C_2$ has no solution if the lines are parallel and distinct, meaning their slopes are equal but their y-intercepts are different: $\frac{A_1}{A_2} = \frac{B_1}{B_2} eq \frac{C_1}{C_2}$ Equating the ratios of the coefficients of x and y: $\frac{k}{3} = \frac{12}{k}$ Cross-multiplying gives: $k^2 = 36 \Rightarrow k = 6$ or $k = -6$ Now we test both values against the constant terms: 1. If $k = 6$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Exponents & Polynomials

Consider the system of linear equations: x Which of the following points is the solution to this system?

a. $(1, 3)$ ✓

b. $(2, 5)$

c. $(3, 7)$

d. $(0, 1)$

To find the solution to the system of equations, we can set the expressions for y equal to each other: $2x + 1 = -x + 4$ Add x to both sides: $3x + 1 = 4$ Subtract 1 from both sides: $3x = 3$ Divide by 3: $x = 1$ Now, substitute $x = 1$ into either of the original equations to find y. Using the first equation: $y = 2(1) + 1$ $y = 2 + 1$ $y = 3$ Therefore, the solution to the system is the point $(1, 3)$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Functions & Transformations

How many solutions does a consistent independent system of two linear equations have?

a. **Exactly one** ✓

b. None

c. Two

d. Infinitely many

Consistent independent ⇒ lines intersect once.

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